METRIC

MIL-C-49310(CR)

17 May 1989

MILITARY SPECIFICATION

COOLER-DEWAR GROUP, OR-338/V, OR-339/V

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers only the cooler-Dewar group to be used in the M1 Thermal Imaging System (TIS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-C-49175	- Cooler, Cryogenic, Mechanical HD-1033B/UA and HD-1033C/UA
MIL-D-49297	- Detector-Dewar DT-636/U

STANDARDS

MILITARY

MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-2073-2	- Packaging Requirement Codes

Beneficial comments (recommendations, additions,deletions) and any pertinent data which may be of use in improving this document should be addressed to:HQ, USA Communications-Electronics Command ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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HANDBOOKS

MIL-HDBK-742	- Waste Disposal Methods for Military Packaging Materials

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other government documents and drawings. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

PURCHASE DESCRIPTION

PD-0182-001A(CR)	- Cooler, Cryogenic Long Life, One Watt Integral HD-1033D/UA

DRAWINGS

54490	5008606	- Test Fixture Assembly Microphonics Standard
54490	5008608	- Mounting Requirements
54490	5008645	- Cooler-Dewar Subassembly, 120 Element

(Copies of Specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item Definition. The cooler-Dewar group is a component of the Thermal Imaging Systems (FLIR) used in the M1. The group consists of a 120 element infrared detector DT-636/U, one watt cryogenic cooler, (HD-1033B/UA, HD-1033C/UA or HD-1033D/UA), a thermal conductor and associated fastening hardware.

3.1.1 Common module selection. The detector and cooler shall meet all requirements of their respective specifications and be inspected in accordance with the Quality Assurance Provisions of these specifications prior to assembly of the cooler-Dewar group, OR-338/V,OR-339/V. The group shall be constructed in accordance with drawing 5008645.

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Detector-Dewar	DT-636/U	MIL-D-49297(CR)
Cooler, Cryogenic, Mechanical	HD-1033B/UA OR HD-1003C/UA	MIL-C-49175B(CR)
OR		
Cooler, Cryogenic Long Life, One Watt Integral	HD-1033D/UA	PD-0182-001A(CR)

3.1.2 Thermal conductor. The bellows is a separate metal piece that thermally connects the cooler cold finger to the detector-Dewar. The bellows shall be installed in accordance with 5008645. The installation procedure of the thermal conductor shall be approved by the Government. (See 6.2.m)

3.2 Parts, materials and processes, general. Parts, materials and processes shall be in accordance with the applicable drawings. Parts, materials and processes not specified shall conform to the requirements of MIL-P-11268.

3.2.1 Construction. The cooler-Dewar group shall be constructed in accordance with 5008645.

3.3 First Article. When specified in the contract the contractor shall furnish first article assemblies in accordance with 4.3.

3.4 Performance requirements. The cooler-Dewar group shall meet the following performance characteristics.

3.4.1 Cooler-Dewar group defective elements. Unless otherwise specified herein, the detector-Dewar shall be operated at 80K (+5, -20K). The detector-Dewar shall contain no more than 12 defective elements as defined in 6.3. Of the 12 defective elements, no more than 5 may be hard defects. In the central 18 elements of the array there shall be no hard defects and no more than 2 soft defects. The central 36 elements shall contain no hard defects and no more than 4 soft defects. Hard defects shall be separated from one another by at least 4 non defective elements and shall be electrically strapped to an adjacent element with common ground. Soft defects located within the central 36 elements shall be separated from any other defect by at least 4 nondefective channels and shall not be strapped unless the strapping is from an adjacent hard defect at the end of a group of 5 commonly grounded elements. (see 6.3.5)

3.4.2 Microphonics. When tested in accordance with paragraph 4.6.2, no elements of the detector array shall have any noise peaks greater than 6dB above the white noise reference value from 300 to 10,000 Hz and 10dB from 10,001 to 20,000 Hz.

3.4.3 Cooldown time. The cooldown time to reach a detector temperature of 80K shall be in accordance with Figure 1.

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3.4.4 Cooler input power. The total input power to the cooler shall not exceed that shown in Figure 2 after detector temperature has reached at least 80K. The input voltage shall be 117 ± 2, volts alternating current (VAC), 400 ± 20 hertz (Hz) power source.

3.5 Environmental conditions.

3.5.1 Temperature extremes. The cooler-Dewar group shall not be damaged when exposed to temperatures from -32°C to +60°C, and stored at temperature from -54°C to +71°C (see 6.3).

3.5.2 Vibration exposure. The cooler-Dewar group shall not be damaged by vibration to the following random and superimposed sinusoidal levels for a period of 15 minutes in each of 3 mutually perpendicular axes.

Random: 1 x 10^{-3} g^2/Hz, 5 to 500 Hz

Sine: 0.26 g's, Octave 1 42 to 84 Hz X axis (cooler piston axis)
0.78 g's, Octave 1 42 to 84 Hz Y axis (cooler motor axis)
0.85 g's, Octave 1 42 to 84 Hz Z axis (cooler coldfinger axis)

3.5.3 Temperature Shock. The cooler-Dewar assembly shall not be damaged (see 6.3.) by sudden changes in temperature between -54°C and +71°C.

3.6 Workmanship. All components and parts shall be manufactured and assembled to insure uniform quality and free from defects. Workmanship shall conform to all applicable portions of the equipment drawings and requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth where such inspections are deemed necessary to meet prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of maufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspections (see 4.5)

4.3 First article inspection. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor. Reliability testing is required only on the cooler for first article testing.

4.3.1 Inspections. All materials, parts, processes and assemblies shall be examined for conformance to the applicable specification or drawing.

TABLE I. Inspections.

Inspection	Requirement paragraph	Inspection Criteria
Any part or component missing or damaged	3.2	MIL-STD-252
Workmanship not as specified	3.6	MIL-STD-454

4.3.2 Tests. Upon successful completion of the inspections specified in 4.3.1, four first article assemblies shall be subjected to all tests listed in Table II. The four first article coolers must sucessfully meet the requirement of the respective cooler specification. Tests may be conducted in any order. Failure of any test shall constitute first article failure. (See 6.2)

4.3.3 Disposition of first article samples. Cooler-Dewar groups which have been subjected to and passed first article inspections may be accepted on the contract or purchase order provided all visible and known damage is repaired and the assemblies are resubjected to and pass Group A inspections. The four reliability first article coolers shall not be considered as part of the procurement quantities.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Defective elements	3.4.1	4.6.1
Detector microphonics	3.4.2	4.6.2
Cooldown time	3.4.3	4.6.3
Cooler input power	3.4.4	4.6.4
Environmental conditions	3.5	4.7

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4.4 Inspection procedures for quality assurance provisions. Unless otherwise specified herein, the cooler-Dewar group shall meet specified performance requirements when operated at an ambient temperature at +23°C ± 5°C with detector temperature of 80K + 5K, -20K, and detector bias of +5.0 Vdc ± 0.2 Vdc and cooler power of 117 ± 2 Vac, 400 Hz ± 20 Hz.

4.5 Quality conformance inspection.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.2 Tests.

4.5.2.1 Group A inspection. Group A inspection shall be conducted on all cooler-Dewar groups. Group A tests listed in Table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A Inspection.

Inspection	Requirement paragraph	Test paragraph
Defective elements	3.4.1	4.6.1
Detector microphonics	3.4.2	4.6.2
Cooldown time	3.4.3	4.6.3
Cooler input power	3.4.4	4.6.4

4.5.2.2 Group B inspection. Not required.

4.5.2.3 Group C inspection. Group C inspection shall be conducted on cooler-Dewar groups selected from units which have passed the tests in 4.5.2.1. The environmental tests of 4.7 shall be performed on a sample quantity defined in 4.5.2.3.1.

4.5.2.3.1 Sampling for group C inspection. Unless otherwise specified, the sample quantity shall be as defined below. The samples shall be randomly selected from units produced during the sampling period.

MONTHLY LOT SIZE	NUMBER OF UNITS MONTHLY
1-9	1
10-29	2
30-up	3

4.5.2.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2 g).

4.5.2.3.3 Reinspection of conforming group C samples. Unless otherwise specified, sample units which have been subjected to and passed Group C inspections may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspections.

4.5.2.4 Group D inspection. Not required.

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4.6 Test requirements. Unless otherwise specified, the group shall be interconnected and operated under standard ambient conditions (temperatures of 23°C ± 5°C and humidity not to exceed 90 percent). The detector shall be operated at +5.0 Vdc ± 0.2 Vdc. Each detector element shall be biased with current flow in the same direction. The cooler shall operate from a 117 Vac ± 2 Vac, 400 Hz ± 20 Hz power source.

4.6.1 Detector-dewar defective elements. The detector array shall be checked for continuity using an ohm meter (current limited to 1 mA). Lack of continuity indicates an open element and shall constitute failure of the tests. The Group A inspection test report on the detector shall be used to verify other requirements of paragraph 3.4.1.

4.6.2 Detector microphonics. The cooler-Dewar group shall be mounted to a microphonics test fixture, defined in drawing 5008606, in accordance with the mounting requirements defined in drawing 5008608. All screw/bolt heads shall be torqued in accordance with the requirements specified in drawing 5008606 and 5008608 prior to the start of the test. The cooler shall be activated and run throughout the duration of the microphonics test procedure. The detector shall be biased in accordance with the requirements of paragraph 3.6.10 of MIL-D-49297(CR). When the detector array reaches an operational temperature between 60 and 80K, the frequency spectral density of each detector element shall be measured using a noise power spectrum analyzer. The spectrum analyzer and any associated electronics/software shall be configured and operated to provide a frequency span of 0-20,000 Hz, a measurement window (filter) in the Flat Top configuration, and a Noise Equivalent Bandwidth of 360 Hz ± 10 Hz. A minimum of 64 samples of the frequency spectrum shall be taken and averaged. The white noise reference level of a given detector element shall be determined by averaging the noise on that element from 12,000 to 20,000 Hz with the cooler power applied. Any detector element which fails to meet the requirements of paragraph 3.4.2 shall be considered a failure of this test.

4.6.3 Cooldown time. The required operating temperature shall be determined by measuring the voltage across the temperature sensor #1 installed in the detector-Dewar with a 1 mA ± .01 mA bias applied. The detector-Dewar shall be operated with bias voltage applied to all sample elements from cooler start up to when the above measurements are made. Failure to meet the requirements of 3.4.3 shall constitute failure of this test.

4.6.4 Cooler input power. Input power shall be measured using on AC wattmeter when the temperature sensor is equal to or less than 80K. Failure to meet the requirements of 3.4.4 shall constitute failure of this test.

4.7 Environmental tests. The following tests shall be performed in the sequence presented in accordance with the General Requirements of MIL-STD-810. During the high and low operating temperature tests, the performance test of 3.4.1, 3.4.3 and 3.4.4 shall be accomplished and shall be successfully completed. The performance tests of Table III shall be performed as a baseline before and after each of the following

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tests. Failure to meet the requirements of 3.4.1, 3.4.2, 3.4.3 or 3.4.4 after each environmental test shall constitute failure of that test. During vibration tests and performance tests 3.4.3 shall be performed and successfully completed.

4.7.1 High temperature. The cooler-Dewar group shall be tested in accordance with MIL-STD-810, Method 501.1, Procedure II, except that operating and +71°C for storage.

4.7.2 Low temperature. The cooler-Dewar group shall be tested in accordance with MIL-STD-810, Method 502.1, Procedure I, except that step 2 shall be -54°C and step 4 shall be -32°C.

4.7.3 Vibration. The cooler-Dewar group shall be tested in accordance with MIL-STD-810, Method 514.2 Procedure VIII and procedures as modified herein. Mount the cooler-Dewar group on the vibration table. With the system operating, the cooler-Dewar group shall be vibrated in each of 3 mutally perpendicular axes to the random and superimposed sinusoidal levels specified in paragraph 3.5.2 for 15 minutes. Paragraph 4.6.3 shall be accomplished during each axis. The appropriate periodic component will be swept up each octave in one minute and down the octave the next minute. Failure to meet the requirements of 3.4.3 during vibration or paragraph 3.4.1, 3.4.2, 3.4.3 or 3.4.4 after vibration shall constitute failure of this test.

4.7.4 Temperature shock. Temperature shock shall be in accordance with MIL-STD-810, Method 503.1, Procedure I, except that the temperature of step 2 shall be -54°C and the temperature for step 1 and 4 shall be +71°C. Failure to meet the requirements of Table III shall constitute failure of this test.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2).

5.2 Preservation. Preservation shall be as specified in MIL-STD-2073-2, Tables I through VII, coded as follows: 2M-1-00-00-NR-X-ED.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cooler-Dewar group is intended for use in the M1 TIS.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (See 2.1.1).

c. Necessary actions by the contractor in the event of a Group C failure.

d. If rough handling and functional tests are required.

e. Levels of preservation and packing (see section 5).

f. Whether first article requirements for the cooler are required for assemblies bought under this specification.

g. Production delivery schedule.

(1) Defined in terms of monthly lots.

(2) Actions required relative to Group C.

(3) Disposition/refurbishment of Group C .

h. MIL-STD-810C shall be used for environmental test of 4.7.

i. The Contractor must order Group A detector inspection reports for use in validation of the requirements of paragraphs 3.4.1 and 4.6.1; Defective Elements. The Contractor must also validate conformance with Group A, C and D inspections for both the Detector-Dewar and the Cooler, Cryogenic, Long Life or Cooler, Cryogenic, Mechanical per paragraph 3.1.1, Common Module selection.

j. Specify drawing 5008645-1 for HD-1033B/UA, HD-1033C/UA cooler or 5008645-2 for HD-1033D/UA cooler.

k. When first article rough handling tests are not required.

l. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

m. Thermal Conductor Assembly procedure shall be submitted to:
Director
CECOM Center for Night Vision and Electro-Optics
ATTN: AMSEL-RD-NV-TS
Fort Belvoir, VA 22060-5677

6.3 Definitions.

6.3.1 Damage. Any breakage, loosening, shifting, evidence of corrosion, failure of any finish, hardware, connection or components, or any deterioration in quality or failure to operate within the specification requirements.

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6.3.2 Defective element. A defective element shall be defined as one which has failed the requirements of 3.6.5, 3.6.5.1, 3.6.6, 3.6.7 and 3.6.14 of MIL-D-49297(CR). A defective element may be classified as either hard or soft. A hard defect is an element which fails the requirements 3.6.5.1, 3.6.6, 3.6.7 or 3.6.14 of MIL-D-49297(CR) or which has a 500K blackbody specific detectivity less than 6.0×10^9 cm $Hz^{\frac{1}{2}}W^{-1}$. Two elements shorted together are considered one hard defect. A soft defect is one which meets all requirements of 3.6.6, 3.6.7 and 3.6.14 of MIL-D-49297(CR) and has a 500K blackbody specific detectivity greater than or equal to 6.0×10^9 cm $Hz^{\frac{1}{2}}W^{-1}$, but less than 8.5×10^9 cm $Hz^{\frac{1}{2}}W^{-1}$.

6.3.3 Element. An element is defined as one channel in the 120-element detector array.

6.3.4 Inspection lot size. An inspection lot is defined as three months continuous production or 120 units, whichever comes first, of those assemblies which have successfully passed Group A inspection.

6.3.5 Strapping. Strapping is defined as electrically connecting a defective element to a nondefective element. The pair of strapped elements thus formed, exhibiting 100 percent crosstalk, shall be considered as a single defect.

6.4 Subject term (keyword) listing.

Bellows
Cooler
Dewar

6.5 Environmental pollution preventive. Disposal methods of packaging materials, to prevent environmental pollution, are contained in MIL-HDBK-742, the material specification, or may be obtained from the preparing activity of the applicable material specification.

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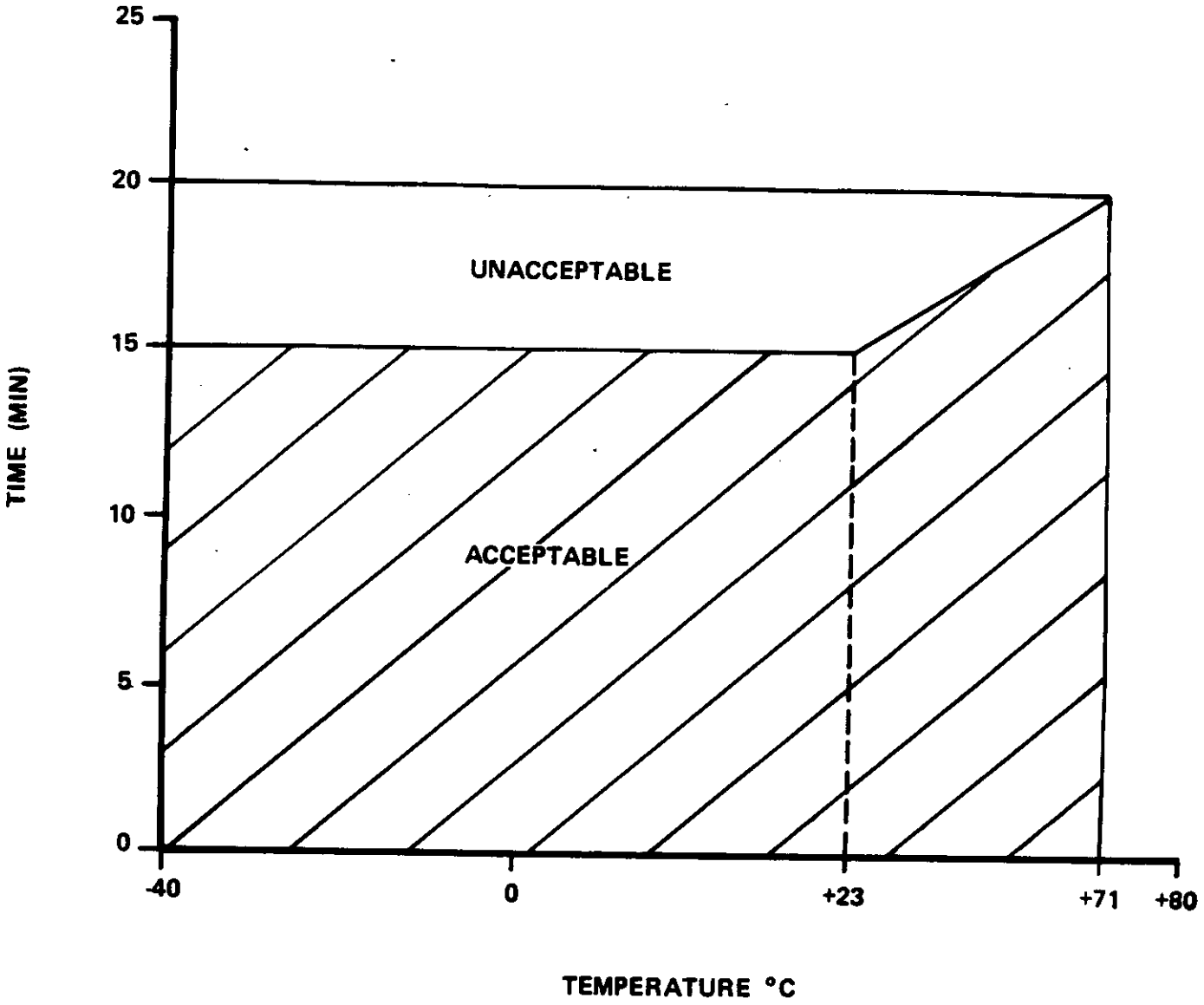


FIGURE 1 Cool down curve.

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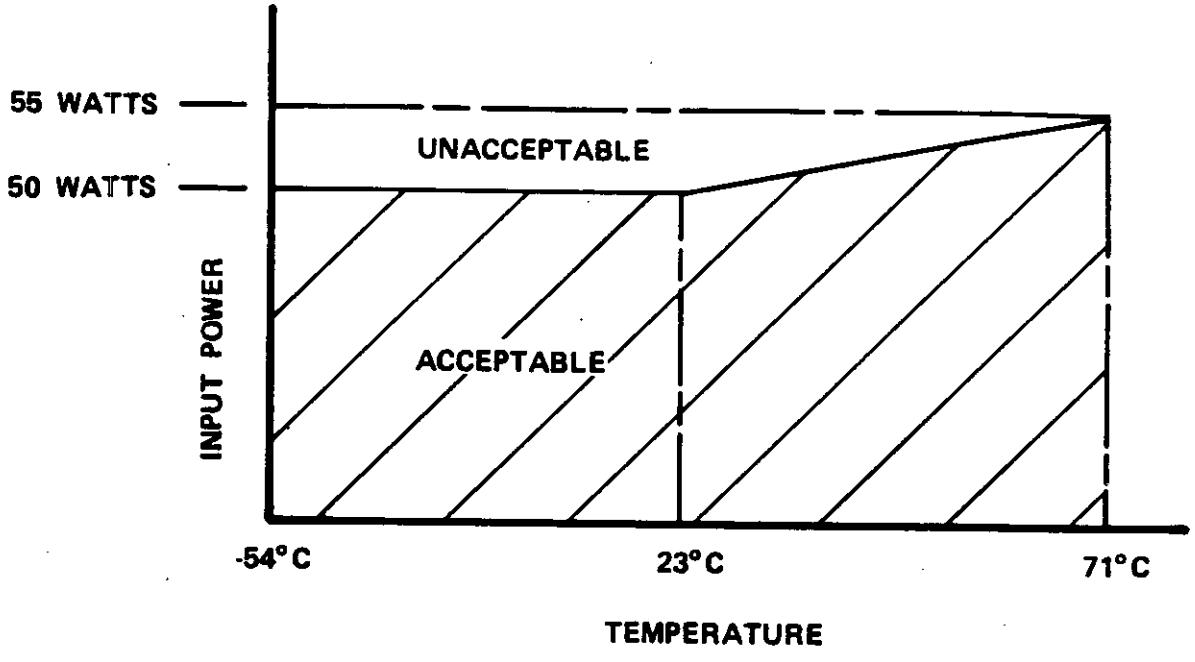


FIGURE 2. Input power with heat load applied.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-49310(CR)	Cooler-Dewar Group, OR-338/V, OR-339/V

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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